CAMBRIDGE IGCSE CHEMISTRY 0620

Topic 4: Electrochemistry

Premium Answer Booklet

18 questions	2020-2025 Paper 4	Premium readable layout

Rebuilt for easier reading with structured tables, clear answer sections and highlighted workings. Use beside the matching topical question bank.

Contents

Question 01 — 19 marks

2020 Feb/March Paper 42 Question 2

Main topic: Topic 4: Electrochemistry
Answer source label: 0620_m20_ms_42.pdf

2(a)(i)	• bauxite
2(a)(ii)	• breakdown by (the passage of) electricity (1) • of an ionic compound in molten / aqueous (state) (1)
2(b)(i)	• cryolite
2(b)(ii)	• less CO2 emission
2(b)(iii)	• Al3+ + 3e- -> Al • any positive Al species gaining electron(s) (1) • correct species and balance (1) • 2O2- -> O2 + 4e- • any negative O species losing electron(s) (1) • correct species and balance (1)
2(b)(iv)	• anodes or carbon / graphite react with oxygen / O2 (1) • (form) carbon dioxide (1)
2(c)(i)	• amphoteric
2(c)(ii)	• aluminium sulfate (1) Al2(SO4)3 (1)
2(c)(iii)	• water
2(c)(iv)	• 2Al(OH)3 -> Al2O3 + 3H2O • species (1) balance (1)
2(c)(v)	• aluminium carbonate (1) • aluminium nitrate (1)

Easy explanation / reminder
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 02 — 8 marks

2020 May/June Paper 42 Question 5

Main topic: Topic 4: Electrochemistry
Answer source label: 0620_s20_ms_42.pdf

5(a)	• breakdown of an ionic compound when molten or in aqueous solution (1) • (using) electricity / electric current / electrical energy (1)
5(b)	• platinum / graphite
5(c)	• $2H^+ + 2e \rightarrow H_2$
5(d)	• Na^+ H^+ Cl^- OH^- • all four (2) 3 or 2 (1)
5(e)	• H^+ and Cl^- are discharged / removed (1) • Na^+ and OH^- remain (1)

Easy explanation / reminder
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 03

14 marks

2020 May/June Paper 43 Question 7

Main topic: Topic 4: Electrochemistry
Cross-topic links: Topic 9: Metals
Answer source label: 0620_s20_ms_43.pdf

7(a)	• breakdown of a molten / or aqueous ionic compound by the passage of electricity
7(b)	• bauxite
7(c)(i)	• it is above carbon in the reactivity series / more reactive than carbon
7(c)(ii)	• any one from: • aluminium oxide has high melting point / cryolite has lower melting point than aluminium oxide • using cryolite reduces costs / expensive to melt aluminium
7(c)(iii)	• oxygen
7(c)(iv)	• Al 3+ + 3e - -> Al
7(d)	• any two related to use as electricity cables: • ductile / malleable • conducts (electricity) • low density • protective oxide layer
7(e)(i)	• iron + water + oxygen ->(hydrated) iron oxide
7(e)(ii)	• any two from: • act as catalysts • variable oxidation numbers • form coloured compounds / coloured ions • higher melting point • higher density • harder

Easy explanation / reminder
• Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
• activation energy.
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

END OF ANSWER

Question 04

15 marks

2020 Oct/Nov Paper 41 Question 5

Main topic: Topic 4: Electrochemistry
Cross-topic links: Topic 7: Acids, bases and salts
Answer source label: 0620_w20_ms_41.pdf

5(a)(i)	• breakdown by (the passage of) electricity (1) • of an ionic compound in molten/aqueous (state) (1)
5(a)(ii)	• they do not react
5(a)(iii)	• negative electrode: hydrogen (gas) (1) • positive electrode: oxygen (gas) (1)
5(a)(iv)	• H+ + e(-) as the only species on the left (1) • equation fully correct (1) • 2H+ + 2e(-) -> H2 (scores 2)
5(b)(i)	• S + O2 -> SO2
5(b)(ii)	• rate of forward reaction is equal to rate of reverse reaction (1) • constant concentration (of reactants and products) (1)
5(b)(iii)	• exothermic / heat / energy is released / surroundings warm up • products have lower energy than reactants / opposite reasoning also allowed
5(c)	• water / H2O
5(d)	• (Mr =) 98 • (0.75 x 98 =) 73.5

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
• escape.

END OF ANSWER

Question 05

19 marks

2021 May/June Paper 41 Question 3

Main topic: Topic 4: Electrochemistry
Answer source label: 0620_s21_ms_41.pdf

3(a)	• 2K + Cl2 -> 2KCl • Cl2 on left hand side (1) • equation fully correct (1)
3(b)	• K outer shell with 8 crosses (1) • Cl outer shell with 7 dots and 1 cross (1) • + and - (1)
3(c)(i)	• breakdown by (the passage of) electricity (1) • of an ionic compound in molten or aqueous (state) (1)
3(c)(ii)	• (anode) chlorine • (cathode)potassium
3(d)(i)	• 2H+ + 2e(-) -> H2 • H+ and e(-) on left hand side (1) • equation fully correct (1)
3(d)(ii)	• chlorine
3(d)(iii)	• potassium hydroxide (1)
3(e)	• one shared pair of electrons and 6 non-bonding electrons on each chlorine atom
3(f)(i)	• liquid (1) • BOTH melting point is below -75 oC AND boiling point is above -75 oC • OR • BOTH -75 oC is higher than -101 oC / melting point AND lower than -35 oC / boiling point • OR • -75 oC is between melting point or -101 oC and boiling point or -35 oC
3(f)(ii)	• ionic bonds in KCl (1) • attraction between molecules in Cl2 (1) • weaker attraction (between particles) in Cl2 opposite reasoning also allowed (1)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

END OF ANSWER

Question 06

12 marks

2021 May/June Paper 42 Question 4

Main topic: Topic 4: Electrochemistry
Answer source label: 0620_s21_ms_42.pdf

4(a)	• electrolyte
4(b)(i)	• 4OH- -> 2H2O + O2 + 4e- • balance of charge (1) • rest of equation (1)
4(b)(ii)	• (OH-(aq) ions) lose electrons
4(c)	• fizzing
4(d)	• 2H+ + 2e- -> H2 • species correct (1) • fully correct equation (1)
4(e)(i)	• fizzing (1) • green gas (1)
4(e)(ii)	• (litmus turns) blue and alkali / base forms (1) • Sodium hydroxide / NaOH (forming) (1)
4(f)	• platinum

Easy explanation / reminder
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 07 — 18 marks

2021 May/June Paper 43 Question 3

Main topic: Topic 4: Electrochemistry
Answer source label: 0620_s21_ms_43.pdf

3(a)	• 2Na + F2 -> 2NaF • F2 (1) • equation fully correct (1)
3(b)	• Na outer shell with 8 crosses (1) • F outer shell with 7 dots and 1 cross (1) • Na+ and F- (1)
3(c)(i)	• breakdown by (the passage of) electricity (1) • of an ionic compound in molten / aqueous (state) (1) • Question Answer
3(c)(ii)	• oxygen • hydrogen
3(d)(i)	• (anode) fluorine (1) • (cathode) sodium (1)
3(d)(ii)	• Na+ + e(-) -> Na
3(e)	• one shared pair of electrons and 6 non-bonding electrons on each fluorine atom
3(f)(i)	• liquid (1) • BOTH melting point is below -195 °C and boiling point is above -195 °C • OR -195 °C is in between melting point and boiling point / -220 °C and -188 °C (1) • OR • BOTH -195 °C is higher than -220 °C / melting point AND lower than -188 °C / boiling point
3(f)(ii)	• ionic bonds in NaF (1) • attraction between molecules in F2 (1) • weaker attraction (between particles) in F2 opposite reasoning also allowed (1) • Question Answer

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

END OF ANSWER

Question 08

13 marks

2021 Oct/Nov Paper 41 Question 3

Main topic: Topic 4: Electrochemistry
Answer source label: 0620_w21_ms_41.pdf

3(a)(i)	• 2Cl - -> Cl 2 + 2e- • Cl 2 (1) • rest of equation (1)
3(a)(ii)	• Oxidation AND lose electrons
3(b)	• effervescence (of colourless gas)
3(c)	• 2H+ + 2e- -> H2 • H+ + e as only species on LHS (1) • rest of equation fully correct (1)
3(d)(i)	• No separate point listed.
3(d)(ii)	• increase (1) • H+ ions being removed (1)
3(e)	• cathode: silver / grey solid (1) • anode: bubbles of orange / brown gas (1)
3(f)	• inert (1) • conducts electricity (1)

Easy explanation / reminder
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 09

12 marks

2021 Oct/Nov Paper 43 Question 2

Main topic: Topic 4: Electrochemistry
Answer source label: 0620_w21_ms_43.pdf

2(a)	• ionic compound • AND either molten or aqueous(or both)(1) • conducts electricity / undergoes electrolysis(1)
2(b)(i)	• oxygen (1) • pink / brown solid (1) • copper (1) • orange / brown / yellow liquid (1) • bromine (1)
2(b)(ii)	• 2H+ + 2e(-) -> H2 (2)
2(b)(iii)	• inert (1) • good conductor of electricity (1)
2(b)(iv)	• electron

Easy explanation / reminder
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 10 — 13 marks

2022 Feb/March Paper 42 Question 4

Main topic: Topic 4: Electrochemistry
Cross-topic links: Topic 9: Metals
Answer source label: 0620_m22_ms_42.pdf

4(a)(i)	• arrow going from Zn to Fe
4(a)(ii)	• Zn -> Zn2+ + 2e- • Zn as only reactant and Zn2+ as only product • correct equation
4(b)(i)	• any metal above zinc in reactivity series
4(b)(ii)	• any metal below iron in reactivity series
4(c)(i)	• hydrogen and oxygen
4(c)(ii)	• water
4(d)(i)	• electrolysis
4(d)(ii)	• mobile ions
4(e)(i)	• hydrogen • chlorine • sodium hydroxide
4(e)(ii)	• sodium • February/March 2022

Easy explanation / reminder
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

END OF ANSWER

Question 11 — 18 marks

2022 May/June Paper 43 Question 5

Main topic: Topic 4: Electrochemistry
Cross-topic links: Topic 3: Stoichiometry, Topic 7: Acids, bases and salts
Answer source label: 0620_s22_ms_43.pdf

5(a)	• (lattice of) positive ions (1) • sea of / delocalised / mobile electrons (1) • attraction between positive ions and electrons (1)
5(b)	• copper (1) • spoon (1) • (aqueous or solution) of named copper salt (1)
5(c)	• (0.2 -> 50 / 1000) = 0.01 (1) • 0.01 (1) • (250 -> 0.01) = 2.5 (1)
5(d)	• solid undissolved
5(e)	• particles have less energy • fewer collisions (between particles) occur per second / per unit time • a smaller percentage / proportion / fraction of collisions (of particles) are successful / have energy above activation • energy / have energy equal to activation energy
5(f)	• copper(II) carbonate OR copper(II) hydroxide
5(g)	• filtration
5(h)	• (a solution that) can dissolve no more solute • at a given temperature
5(i)	• forms anhydrous (copper sulfate) • OR forms (white) powder

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
• activation energy.

END OF ANSWER

Question 12 — 15 marks

2022 Oct/Nov Paper 43 Question 3

Main topic: Topic 4: Electrochemistry
Answer source label: 0620_w22_ms_43.pdf

3(a)	• bauxite
3(b)	• breakdown by (the passage of) electricity (1) • of an ionic compound in molten or aqueous (state) (1)
3(c)(i)	• improves conductivity of the electrolyte / makes the electrolyte a better conductor (1) • lower operating temperature (1)
3(c)(ii)	• $Al^{3+} + 3e^- \rightarrow Al$ • only Al^{3+} + (3) e^- on the left (1) • equation fully correct (1)
3(c)(iii)	• anodes or carbon react(s) with oxygen (1) • form carbon dioxide (1)
3(d)	• unreactive coating of aluminium oxide (1)
3(e)(i)	• neutralises both acids AND alkalis
3(e)(ii)	• $2NaOH + Al_2O_3 \rightarrow 2NaAlO_2 + H_2O$ • $NaAlO_2$ on the right hand side (1) • equation fully correct (1)
3(f)	• $GaCl_3$ (1) • $Ga_2(SO_4)_3$ (1)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

END OF ANSWER

Question 13

16 marks

2023 May/June Paper 41 Question 5

Main topic: Topic 4: Electrochemistry
Answer source label: 0620_s23_ms_41.pdf

5(a)(i)	• breakdown by (the passage of) electricity(1) • of an ionic compound in molten or aqueous (state) (1)
5(a)(ii)	• graphite is inert AND graphite conducts electricity
5(a)(iii)	• 2H+ + 2e- -> H2 • H+ + e as only species on LHS(1) • equation correct(1)
5(a)(iv)	• electrons
5(a)(v)	• ions
5(a)(vi)	• oxygen(1) • hydrogen(1)
5(b)(i)	• aluminium oxide
5(b)(ii)	• any two from: • ■■ • solvent • ■■ • lowers the operating temperature • ■■ • increases conductivity
5(b)(iii)	• carbon reacts with oxygen and forms carbon dioxide
5(c)(i)	• 2H2 + O2 -> 2H2O • all formulae(1) equation correct(1)
5(c)(ii)	• no carbon dioxide evolved • OR • more efficient

Easy explanation / reminder
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 14

16 marks

2023 May/June Paper 43 Question 5

Main topic: Topic 4: Electrochemistry
Answer source label: 0620_s23_ms_43.pdf

5(a)(i)	• ionic compound • molten and / or aqueous
5(a)(ii)	• oxidation number (of copper)
5(a)(iii)	• fades / (becomes) colourless
5(a)(iv)	• Cu2+ + 2e -> Cu • Cu2+ and (any number of) e on left hand side • equation correct
5(a)(v)	• OH-
5(b)	• anode dissolves
5(c)(i)	• silver • spoon • (aqueous or solution) of silver nitrate
5(c)(ii)	• prevent corrosion • improve appearance
5(d)(i)	• carbon dioxide: (increased) global warming • carbon monoxide: toxic
5(d)(ii)	• needs high pressure to store hydrogen

Easy explanation / reminder
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
• Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
• reduced by carbon/CO.

END OF ANSWER

Question 15

13 marks

2024 Oct/Nov Paper 42 Question 2

Main topic: Topic 4: Electrochemistry
Answer source label: 0620_w24_ms_42.pdf

2(a)	• bauxite
2(b)	• cryolite
2(c)	• (it has) mobile ions
2(d)(i)	• 4 • Al 3+ + 3e- -> Al • Al 3+ AND e- • correct equation
2(d)(ii)	• electrons are lost (from oxide ions)
2(d)(iii)	• anode / electrode made of carbon or graphite • anode / electrode reacts with oxygen / burns / combusts (in oxygen)
2(e)	• aluminium is resistant to corrosion
2(f)	• eight crosses in second shell of Al • seven dots and one cross in second shell of each F • ‘3+' charge on Al ion on correct answer line AND ‘-‘ charge on each F ion on correct answer line

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

END OF ANSWER

Question 16 | 11 marks

2024 Oct/Nov Paper 43 Question 2

Main topic: Topic 4: Electrochemistry
Answer source label: 0620_w24_ms_43.pdf

2(a)	• breakdown by (the passage of) electricity • of an ionic compound in molten or aqueous (state)
2(b)(i)	• brown solution OR black solid • iodine • hydrogen • copper
2(b)(ii)	• 4OH- -> 2H2O + O2 + 4e- • OH- AND e- • correct equation
2(c)(i)	• copper OR anode forms Cu2+ • which goes into the solution
2(c)(ii)	• copper
2(c)(iii)	• no change

Easy explanation / reminder
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 17

15 marks

2025 May/June Paper 42 Question 3

Main topic: Topic 4: Electrochemistry
Answer source label: 0620_s25_ms_42.pdf

3(a)	• ionic • molten • in aqueous solution
3(b)(i)	• increase
3(b)(ii)	• becomes paler blue • or • becomes colourless
3(b)(iii)	• bubbles
3(b)(iv)	• 4OH- -> 2H2O + O2 + 4e- • any negatively charged OH species losing electrons • OR • H2O + O2 as only products (other than electrons) on RHS • any negatively charged OH species losing electrons • AND • H2O + O2 as only products (other than electrons) on RHS • correct equation
3(c)(i)	• increases
3(c)(ii)	• no change
3(d)(i)	• bauxite
3(d)(ii)	• cryolite
3(d)(iii)	• anodes or carbon / graphite react with oxygen / O2 • (form) carbon dioxide

Easy explanation / reminder
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 18 — 17 marks

2025 Oct/Nov Paper 43 Question 3

Main topic: Topic 4: Electrochemistry
Cross-topic links: Topic 9: Metals
Answer source label: 0620_w25_ms_43.pdf

3(a)(i)	• aluminium oxide
3(a)(ii)	• cryolite
3(a)(iii)	• carbon
3(a)(iv)	• Al 3+ + 3e- -> Al • only Al 3+ + 3e- on the left • equation completely correct
3(b)(i)	• unreactive coating of aluminium oxide
3(b)(ii)	• 2Al + 3Zn2+ -> 2Al 3+ + 3Zn
3(b)(iii)	• Zn2+ = +2 • Zn = 0 / zero
3(b)(iv)	• decrease in oxidation number
3(c)(i)	• enthalpy change (of reaction)
3(c)(ii)	• (forward) reaction is exothermic
3(d)	• oxides that react with acids AND with bases to produce a salt and water
3(e)	• 2NaOH + Ga2O3 -> 2NaGaO2 + H2O • NaGaO2 on the right-hand side of an equation • equation fully correct
3(f)	• GaBr3 • Ga2(SO4)3

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

END OF ANSWER